

# Dongfang Zhao, Ph.D.

Nationality: United States of America

## Education

|                                           |                           |      |
|-------------------------------------------|---------------------------|------|
| University of Washington, Seattle         | Computer Science, PostDoc | 2017 |
| Illinois Institute of Technology, Chicago | Computer Science, PhD     | 2015 |
| Emory University, Atlanta, GA             | Computer Science, MS      | 2008 |

## Employment

|           |                                                                                                                                                                                                                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023–now  | University of Washington (UW) <ul style="list-style-type: none"><li>◦ Assistant Professor, UW Tacoma School of Engineering &amp; Technology</li><li>◦ Adjunct Professor, Paul G. Allen School of Computer Science &amp; Engineering</li><li>◦ Data Science Affiliate, eScience Institute</li></ul> |
| 2018–2023 | Affiliate Faculty, University of California, Davis                                                                                                                                                                                                                                                 |
| 2018–2023 | Assistant Professor, University of Nevada, Reno (UNR)                                                                                                                                                                                                                                              |
| 2015      | Research Intern, IBM Almaden Research Center, San Jose, CA                                                                                                                                                                                                                                         |
| 2014      | Research Intern, Argonne National Laboratory, Chicago, IL                                                                                                                                                                                                                                          |
| 2013      | Research Intern, Pacific Northwest National Laboratory, Seattle, WA                                                                                                                                                                                                                                |
| 2009–2011 | Software Developer (full-time), Epic Systems Corporation, Madison, WI                                                                                                                                                                                                                              |

## Awards

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|------|----------------------------------------------------------------------------|
| 2022 | Federal Research Grant, UNR PI, NSF 2112345, \$255,916                     |
| 2020 | Federal Research Grant, UNR PI, DOE SC0020455, \$200,000                   |
| 2019 | Best Paper Award, International Conference on Cloud Computing              |
| 2018 | Best Student Paper Award, IEEE International Conference on Cloud Computing |

## Professional Services

|          |                                                                               |
|----------|-------------------------------------------------------------------------------|
| 2018–now | Associate Editor, Journal of Big Data (Springer)                              |
| 2021–now | Editorial Board, IEEE Transactions on Parallel and Distributed Systems (TPDS) |
| 2025–now | Senior Program Committee, AAAI Conference on Artificial Intelligence (AAAI)   |

## Recent Projects

|                     |                                                                                                                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[ElasticIVF]</b> | Elastic in-memory vector indexing and search for streaming data analysis on heterogeneous platforms: <a href="https://github.com/hpdc/ElasticIVF">https://github.com/hpdc/ElasticIVF</a><br><i>Tags:</i> GPUs, streaming data analysis, parallel computing, FAISS, C++, Python, Bash          |
| <b>[AdaDisk]</b>    | Distributed agentic systems for adaptive ingestion-query scheduling of disk-resident vector indices in LLM serving: <a href="https://github.com/hpdc/AdaDisk">https://github.com/hpdc/AdaDisk</a><br><i>Tags:</i> large language models, AI agents, distributed systems, DiskANN, C++, Python |
| <b>[Hermes]</b>     | MySQL plug-in for high-performance homomorphic encryption of vector databases in privacy-preserving RAG: <a href="https://github.com/hpdc/Hermes">https://github.com/hpdc/Hermes</a><br><i>Tags:</i> retrieval-augmented generation, vector databases, homomorphic encryption, C++            |

## Selected Publications (Dongfang Zhao was/is the \*student's academic advisor)

- [1] \*Baigang Chen, **Dongfang Zhao**. Reliable Non-Leveled Homomorphic Encryption for Web Services. The ACM Web Conference (formerly International World Wide Web Conference, WWW), 2026.
- [2] \*Chengyu Gong, \*Gefei Shen, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. Order-Preserving Dimension Reduction for Multimodal Semantic Embedding. The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026.
- [3] \*Jiuzhou Fu, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. ProHD: Projection-Based Hausdorff Distance Approximation. IEEE International Conference on Data Mining (ICDM), 2025.
- [4] \*Jiayao Wang, ..., **Dongfang Zhao**. IPBA: Imperceptible Perturbation Backdoor Attack in Federated Self-Supervised Learning. European Conference on Artificial Intelligence (ECAI), 2025.
- [5] \*Abdullah Al-Mamun, **Dongfang Zhao**, et al. ZTP: A Scalable and Lightweight Privacy-Preserving Blockchain via Scale-Free Quorums and Geometric Fragmentation. International Conference on Parallel Processing (ICPP), 2025.
- [6] \*Gefei Shen, **Dongfang Zhao**. Towards Nonparametric Topological Layers in Neural Networks. The 28th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2024.
- [7] \*Muhammad Jahanzeb Khan, ..., **Dongfang Zhao**. Privacy-Preserving Artificial Intelligence on Edge Devices: A Homomorphic Encryption Approach. IEEE International Conference on Web Services (ICWS) 2024: 395-405.
- [8] \*Chunyu Yuan, **Dongfang Zhao**, Sos S. Agaian. UCM-Net: A lightweight and efficient solution for skin lesion segmentation using MLP and CNN. Biomedical Signal Processing and Control, Volume 96, Part B, 17 pages, 2024.
- [9] \*Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data (SIGMOD), 2023.
- [10] \*Xinying Wang, Lipeng Wan, Scott Klasky, **Dongfang Zhao**, Feng Yan: SciLance: Mitigate Load Imbalance for Parallel Scientific Applications in Cloud Environments. International Conference on Cluster Computing (CLUSTER), 2023.
- [11] \*Olamide Timothy Tawose, Lei Yang, **Dongfang Zhao**. TopoCommit: A Topological Commit Protocol for Cross-Ledger Transactions in Scientific Computing. International Conference on Cluster Computing (CLUSTER), 2023.
- [12] \*Abdullah Al-Mamun, **Dongfang Zhao**. ECHAIN: Securing Electronic Device Provenance through Privacy-Preserving Consortium Blockchains. The 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023.
- [13] **Dongfang Zhao**. Cross-Blockchain Transactions: Systems, Protocols, and Topological Theory. In *Handbook on Blockchain*, pp. 157-193. Cham: Springer International Publishing, 2022.
- [14] \*Xinying Wang, ..., **Dongfang Zhao**. Memory Scaling of Cloud-based Big Data Systems: A Hybrid Approach. IEEE Transactions on Big Data (TBDData). 8(5): 1259-1272 (2022).
- [15] \*Abdullah Al-Mamun, Haoting Shen, **Dongfang Zhao**. DEAN: A Lightweight and Resource-efficient Blockchain Protocol for Reliable Edge Computing. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [16] \*Olamide Timothy Tawose, \*Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.

- [17] \*Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [18] \*Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021.
- [19] \*Xinying Wang, \*Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020.
- [20] \*Abdullah Al-Mamun, ..., **Dongfang Zhao**. Poster Reflector: a fine-grained I/O tracker for HPC systems. ACM Symposium on Principles and Practice of Parallel Programming (PPoPP), 2020.
- [21] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig. Locality-Aware Scheduling for Containers in Cloud Computing. IEEE Transactions on Cloud Computing (TCC). 8(2): 635-646 (2020).
- [22] \*Xinying Wang, \*Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud. The 12th International Conference on Cloud Computing (CLOUD), 2019.
- [23] Heyang Qin, Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan. Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [24] \*Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**. Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory. The 11th IEEE International Conference on Cloud Computing (IEEE CLOUD), 2018.
- [25] \*Jiayao Wang, \*Abdullah Al-Mamun, Tonglin Li, Linhua Jiang, **Dongfang Zhao**. Toward Performant and Energy-efficient Queries in Three-tier Wireless Sensor Networks. Proceedings of the 47th International Conference on Parallel Processing (ICPP), 2018.
- [26] Parmita Mehta, Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad. Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of Very Large Data Bases (VLDB), 2017.
- [27] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu. Toward Efficient and Flexible Metadata Indexing of Big Data Systems. IEEE Transactions on Big Data (TBData). 3(1): 107-117 (2017).
- [28] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Xiaobing Zhou, Ioan Raicu. Exploiting multi-cores for efficient interchange of large messages in distributed systems. Concurrency and Computation: Practice and Experience (CCPE). 28(13): 3568-3585 (2016).
- [29] **Dongfang Zhao**, Ke Wang, Kan Qiao, Tonglin Li, Iman Sadooghi, Ioan Raicu. Toward high-performance key-value stores through GPU encoding and locality-aware encoding. Journal of Parallel and Distributed Computing (JPDC). 96: 27-37 (2016).
- [30] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Services Computing (TSC). 9(1): 96-109 (2016).
- [31] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems (TPDS). 27(6): 1824-1837 (2016).

- [32] **Dongfang Zhao**, NagaPramod Mandagere, Gabriel Alatorre, Mohamed Mohamed, Heiko Ludwig. Toward locality-aware scheduling for containerized cloud services. IEEE International Conference on Big Data (BigData), 2015.
- [33] **Dongfang Zhao**, Kan Qiao, Ioan Raicu. HyCache+: Towards Scalable High-Performance Caching Middleware for Parallel File Systems. The 14th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid), 2014.
- [34] **Dongfang Zhao**, Chen Shou, Tanu Malik, Ioan Raicu. Distributed data provenance for large-scale data-intensive computing. International Conf. on Cluster Computing (CLUSTER), 2013.
- [35] Tonglin Li, Xiaobing Zhou, Kevin Brandstatter, **Dongfang Zhao**, Ke Wang, Anupam Rajendran, Zhao Zhang, Ioan Raicu. ZHT: A Light-Weight Reliable Persistent Dynamic Scalable Zero-Hop Distributed Hash Table. IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2013.
- [36] **Dongfang Zhao**, Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI). 31(1): 86-98 (2009).